Xiang Ye · 叶翔

PhD Candidate in Statistics | KAUST

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Personal Profile

I'm a PhD candidate in the Bayesian Computational Statistics Group in King Abdullah University of Science and Technology (KAUST), supervised by Professor Håvard Rue, and I expect to finish my PhD in Statistics around the end of 2026. I am enthusiastic about Bayesian computational methodologies and latent variable modeling, especially the design of priors and model structures for scalable probabilistic models, and the synergy between Bayesian inference and modern high-dimensional learning algorithms. My interests extend to uncertainty quantification and directional statistics.

Education

King Abdullah University of Science and Technology (KAUST) — Jeddah, Saudi Arabia
PhD in Statistics — August 2023 - December 2026

- Research Group: Bayesian Computational Statistics
- **Thesis:** A Bayesian regression framework for circular models
- **Courses:** Probability and Statistics, Linear Models, Stochastic Processes, Spatial Data Science with R, Numerical Optimization, Functional Data Analysis, Stochastic Partial Differential Equations

Lancaster University — Lancaster, UK
MSc in Statistics — October 2021 - September 2022

- Graduate with distinction
- **Courses:** Statistical Inference, Bayesian Statistics, Statistcal Learning, Survival Analysis, Methods for Missing Data, Computationally Intensive Methods (Markov Chain Monte Carlo, Particle Filter), Time-Series, Multivariate Time-Series, Programming in R, Programming in Python
- **Dissertation:** Spatial Statistical Modeling with INLA (This dissertation is graded as "Distinction")

University of Liverpool & Xi'an Jiaotong-Liverpool University (XJTLU) — Suzhou, China
BSc in Applied Mathematics — September 2017 - June 2021

- Graduated with Upper Second Class
- **Courses:** Calculus, Multivariate Calculus, Linear Algebra, Advanced Linear Algebra, Abstract Algebra, Vector Field Theory, Analysis, Probability and Statistics, Statistical Distribution Theory, Ordinary Differential Equations, Partial Differential Equations, Complex Functions, Dynamic Modeling, Numerical Analysis, Operational Research, Metric Spaces, Population Dynamics, Stochastic Process, Differential Geometry, Optimisation Theory, Quantum Mechanics, Functional Analysis, Programming in JAVA
- **Final Year Project:** How to find the best starting lineup for a basketball team? – An application for integer programming to the management of an NBA team. (This thesis is graded as "Distinction")

Publications

Ye, **X.**, van Niekerk, J., and Rue, H. (2026). A Bayesian regression framework for circular models with INLA. *Journal of the Royal Statistical Society Series C: Applied Statistics*, qlag056. doi:10.1093/jrsssc/qlag056.

Ye, **X.**, van Niekerk, J., and Rue, H. (2026). Penalizing complexity priors for Bayesian inference of circular models. *Statistical Modelling*, 1471082X261453677. doi:10.1177/1471082X261453677.

Honors and Awards

Department Prize for Best Computational Dissertation — Lancaster, UK
Lancaster University — October 2022

- Dissertation: Spatial Statistical Modeling with INLA

Meritorious Winner of Mathematical Contest In Modeling (MCM) — USA
Consortium for Mathematics and its Applications — March 2020

- Meritorious Winner as well as Student Advisor
- Analyze the selling data and customers' review of the products of a company, find patterns and make predictions to give suggestions to help the company to put their product into the Amazon online market.

Excellent Team Member of ENACTUS — Suzhou, China
ENACTUS (Students in Free Enterprise) — September 2017 - June 2019

- Helping to protect and maintain the traditional legacy of nut-cutting handicraft through business operating mode.

Intern and Research Experience

Research and Development Intern — Guangzhou, China
Tech View Info Limited Liability Company — February 2023 - May 2023

- Duties include:
- Combine and analyze the data from many resources, do data engineering to process the data to meet the need of the company.
- Construct and optimize statistical models and machine learning models to do predictions with respect to the data to meet the need of the clients
- Design new algorithms, methodology and systems for constructing machine learning models with respect to the different needs for the company, and write patent applications

Statistical Assistant — Guangzhou, China
National Social Science Foundation of China — March 2020 - August 2020

- Statistical assistant in a nationwide investigation of Teacher Talk project supported by the National Social Science Foundation of China (Grant NO. 16BYY090), providing assistance in data sorting and statistical analysis.

Research assistant — Guangzhou, China
Professor Ronghua Li of Beijing Institute of Technology — August 2019 - August 2020

- Research assistant in the program of PageRank algorithm with Professor Ronghua Li

Project Marketing Intern — Guangzhou, China
Guangdong Zhujiang Investment CO LTD — July 2019 - August 2019

- Duties include statistical analysis of data collected by the department using EXCEL and other software to process data and construct charts or graphs, and writing reports.

Skills

Programming	R (INLA, STAN, ggplot2, caret, keras), Python, C++, MATLAB
Tools & Software	LaTeX, Microsoft Office, Adobe Photoshop, Adobe Premier Pro

Interests

Sports	I love sports and good at many ball games. I love football and basketball and I am good at table tennis.
Arts	I love drawing. I have won the first class prize at a National Drawing Contest for Middle School Students of China in 2011.
Music	I enjoy music. I have learnt violin since primary school. I have also learnt piano and saxophone.
Reading	I love reading very much. Two of my favorite books are *The Count of Monte Cristo* and *One Hundred Years of Solitude*.
Cooking	I love cooking. I am good at many Chinese dishes an enjoy trying new dishes.

Languages

English	Professional proficiency, IELTS 7 (2021)
Chinese	Native proficiency
German	A2

References available upon request.